Complete Developer Integration Manual & API Reference

Full Technical Specification for Connecting ERP, CRM, POS & Billing Systems with Meta WhatsApp Cloud API

Base Production URL:	**https://whatapp-automation-kxml.onrender.com**
Interactive Hub / Web:	**https://itechmobileswhatsappautomation.vercel.app**
API Authentication:	Header: x-api-key: itech_live_sec_7f9984e1b9384e9d9a1f
Meta WABA Account ID:	1758570912097342 (Official Verified Account)
Supported Standards:	REST / JSON over HTTPS · Meta Cloud API v20.0 · Webhooks
Official Store Address:	TV Malai Road, Tirukannur, Pondicherry - 605501
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1. System Architecture & How ERP Connects

The iTechnic Mobiles WhatsApp Gateway acts as a high-throughput, secure middleware connecting your in-house **Billing Software, ERP, POS, or Inventory Management System** directly to Meta's WhatsApp Cloud API (Graph API v20.0). Whenever a business event occurs in your ERP (such as repair job creation, status progression, invoice billing, or dispatch), your system fires a simple HTTP POST request to this gateway. The gateway validates authentication, automatically formats parameters into the designated pre-approved Meta Utility Template, guarantees 24/7 delivery outside WhatsApp's 24-hour window, and synchronizes the conversation in the centralized MongoDB customer ledger.

2. Production Environment & Security Model

All requests to /api/v1/* require authentication via the production API Secret Key. The key must be transmitted in either the x-api-key header or the Authorization: Bearer header.

```
POST /api/v1/crm/repair-status-update HTTP/1.1
Host: whatapp-automation-kxml.onrender.com
x-api-key: itech_live_sec_7f9984e1b9384e9d9a1f
Content-Type: application/json
Accept: application/json
```

3. Quick Start: 5-Minute Connectivity Test

To verify that your server can authenticate with the gateway, execute a GET call to /api/v1/auth/key:

```
curl -X GET "https://whatapp-automation-kxml.onrender.com/api/v1/auth/key" \
  -H "x-api-key: itech_live_sec_7f9984e1b9384e9d9a1f"

// Response (200 OK):
{
  "success": true,
  "apiKey": "itech_live_sec_7f9984e1b9384e9d9a1f",
  "wabaId": "1758570912097342",
  "baseUrl": "https://whatapp-automation-kxml.onrender.com/api/v1"
}
```

4. CRM Service & Repair Desk APIs

Integrate these 6 core endpoints into your job ticketing, workshop, and billing lifecycle. The gateway normalizes Indian 10-digit phone numbers automatically by adding 91.

4.1 Repair Booking Confirmation

Trigger in ERP: When customer brings or logs a phone for service, screen replacement, or repair.

Endpoint: POST /api/v1/crm/repair-booking-confirmed

```
POST /api/v1/crm/repair-booking-confirmed
{
  "phone": "919876543210",              // Mandatory (with country code, no +)
  "customerName": "Ramesh Babu",        // Mandatory
  "jobId": "REP-2026-001",              // Unique ERP Service Job ID
  "device": "Samsung Galaxy S23",       // Device Name / Model
  "service": "Display Glass & Touch",   // Reported Problem
  "turnaround": "Today by 6:00 PM"      // Expected Completion Time (optional)
}
```

4.2 Live Service Status Update

Trigger in ERP: When technician updates job card (In Repair, Waiting for Spares, Quality Testing).

Endpoint: POST /api/v1/crm/repair-status-update

```
POST /api/v1/crm/repair-status-update
{
  "phone": "919876543210",
  "customerName": "Ramesh Babu",
  "jobId": "REP-2026-001",
  "device": "Samsung Galaxy S23",
  "status": "In Progress - Screen Replaced, Running QA Benchmarks"
}
```

4.3 Device Ready for Pickup & Final Bill Delivery

Trigger in ERP: When device passes QC and invoice/bill is finalized in billing software.

Endpoint: POST /api/v1/crm/device-ready-pickup

```
POST /api/v1/crm/device-ready-pickup
{
  "phone": "919876543210",
  "customerName": "Ramesh Babu",
  "jobId": "REP-2026-001",
  "device": "Samsung Galaxy S23",
  "finalCost": "4500",                  // Total Amount in INR
  "billNo": "INV-89104"                 // Tax Invoice Number
}
```

4.4 Service Cost Quotation & Customer Approval

Trigger in ERP: Post-inspection when customer authorization is required before replacing parts.

Endpoint: POST /api/v1/crm/service-quotation

```
POST /api/v1/crm/service-quotation
{
  "phone": "919876543210",
  "customerName": "Ramesh Babu",
  "jobId": "REP-2026-001",
  "device": "Samsung Galaxy S23",
  "issue": "Motherboard Power IC Replacement",
  "quotedCost": "2800"
}
```

4.5 90-Day Digital Warranty Certificate

Trigger in ERP: Dispatched immediately after payment settlement to guarantee customer trust.

Endpoint: POST /crm/repair/send-warranty

```
POST /crm/repair/send-warranty
{
  "phone": "919876543210",
  "repairId": "REP-2026-001"
}
```

4.6 Doorstep Pickup Scheduled / Field Visit

Trigger in ERP: When home pickup or on-site repair is scheduled.

Endpoint: POST /api/v1/crm/doorstep-pickup

```
POST /api/v1/crm/doorstep-pickup
{
  "phone": "919876543210",
  "customerName": "Ramesh Babu",
  "timeSlot": "Today between 4:00 PM - 5:30 PM",
  "address": "No. 14, Main Road, Tirukannur, Pondicherry"
}
```

5. Retail Commerce, Orders & Leads APIs

Sync products, live stock counts, smartphone customer orders, and sales leads between your ERP and the WhatsApp bot.

5.1 Order Dispatch & Tracking Notification

Endpoint: PUT /api/v1/orders/:id/status & POST /api/v1/orders/:id/send-update

Updates order to *Order Confirmed, Processing, Shipped, Out for Delivery, or Delivered* and sends visual 5-stage timeline to customer WhatsApp.

```
// 1. Update Order in Gateway DB:
PUT /api/v1/orders/ORD-1082/status
{
  "orderStatus": "Out for Delivery",
  "courierPartner": "Dunzo Express / Store Rider",
  "trackingNumber": "TRK-982130",
  "assignedRider": { "name": "Vimal", "phone": "919876543210" }
}

// 2. Dispatch 5-Stage WhatsApp Milestone Message to Customer:
POST /api/v1/orders/ORD-1082/send-update
```

5.2 Warehouse Inventory & Price Sync (ERP → WhatsApp Bot)

Endpoint: PUT /api/v1/products/:id

Automatically updates price and availability in the WhatsApp Mobile Buyer catalog (Pillars 1 & 4).

```
PUT /api/v1/products/phone_apple_16pro
{
  "stock": 14,                          // Current Units in Stock
  "price": 119900,                      // Standard MRP
  "offerPrice": 114900,                 // Special Discount Price
  "inStock": true
}
```

5.3 Sales Leads Ingestion (WhatsApp Bot → ERP CRM)

Endpoint: GET /api/v1/leads & PUT /api/v1/leads/:id/status

Poll high-intent leads generated from WhatsApp phone recommendation wizards and EMI calculators into your sales queue.

```
// GET /api/v1/leads Response:
{
  "success": true,
  "count": 3,
  "leads": [
    {
      "leadId": "LEAD-1002",
      "customerName": "Anand Kumar",
      "customerPhone": "919342400879",
      "targetDevice": "Apple iPhone 16 Pro",
      "budgetRange": "Above ₹80,000",
      "purchasePreference": "Visit Showroom",
      "status": "Assigned",
      "assignedExecutive": { "name": "Karthik Raja", "phone": "919342400879" }
    }
  ]
}

// Update status after sales representative contacts customer:
PUT /api/v1/leads/LEAD-1002/status
{
  "status": "Contacted",
  "noteText": "Customer confirmed visit at 4:30 PM for iPhone 16 Pro Desert Titanium."
}
```

5.4 Bank Offers & Support Agent Presence

Offers: GET /api/v1/offers & PUT /api/v1/offers/:id (Activate/deactivate credit card EMI deals).

Agents: GET /api/v1/agents & PUT /api/v1/agents/:id/status (Toggle online presence for departments).

6. Universal Messaging & Template Control

When you need to trigger a custom Meta template or send a message that doesn't fit standard CRM workflows, use the Universal Messaging endpoints.

6.1 Trigger Any Approved Meta Template

Endpoint: POST /api/v1/messages/send-template

```
POST /api/v1/messages/send-template
{
  "to": "919876543210",
  "templateName": "repair_booking_confirmed",
  "languageCode": "en_US",
  "bodyVariables": ["Ramesh", "REP-1024", "iPhone 13", "Screen", "Same Day"]
}
```

6.2 Send Direct WhatsApp Text (24-Hour Window Only)

Endpoint: POST /api/v1/messages/send-text

```
POST /api/v1/messages/send-text
{
  "to": "919876543210",
  "text": "Hello Ramesh, our technician has completed testing your phone. Everything is perfect!"
}
```

6.3 Template Management (List & Submit to Meta)

List Templates: GET /api/v1/templates (Returns live approval status: APPROVED, PENDING, REJECTED).

Create Template: POST /api/v1/templates (Submits new template directly to Meta for review).

7. Inbound Webhook Ingestion & Event Callbacks

The gateway listens for inbound customer messages, button selections, and delivery status updates at /webhook.

```
// Inbound Webhook Verification (Handled automatically by Gateway):
GET /webhook?hub.mode=subscribe&hub.challenge=1158201444&hub.verify_token=YOUR_TOKEN

// Inbound Message Delivery Status Receipt from Meta:
{
  "entry": [{
    "changes": [{
      "value": {
        "statuses": [{
          "id": "wamid.HBgM0TE5...",
          "status": "delivered",       // sent | delivered | read | failed
          "timestamp": "1710140201",
          "recipient_id": "919876543210"
        }]
      }
    }]
  }]
}
```

8. Multi-Language Code Snippets (5 Stacks)

Production-ready, copy-paste implementations for your development team.

8.1 Python 3 (Requests / Odoo / Django / FastAPI)

```python
import requests

GATEWAY_URL = 'https://whatapp-automation-kxmi.onrender.com/api/v1/crm/device-ready-pickup'
HEADERS = {
    'x-api-key': 'itech_live_sec_7f9084e1b9384e9d9a1f',
    'Content-Type': 'application/json'
}
payload = {
    'phone': '919876543210',
    'customerName': 'Ramesh Babu',
    'jobId': 'REP-2026-081',
    'device': 'Samsung Galaxy S23',
    'finalCost': '4500',
    'billNo': 'INV-89104'
}

response = requests.post(GATEWAY_URL, json=payload, headers=HEADERS, timeout=10)
print('Status:', response.status_code, response.json())
```

8.2 Node.js / JavaScript (Axios / Fetch / TypeScript)

```javascript
const axios = require('axios');

async function notifyDeviceReady() {
  const res = await axios.post(
    'https://whatapp-automation-kxmi.onrender.com/api/v1/crm/device-ready-pickup',
    {
      phone: '919876543210',
      customerName: 'Ramesh Babu',
      jobId: 'REP-2026-081',
      device: 'Samsung Galaxy S23',
      finalCost: '4500',
      billNo: 'INV-89104'
    },
    { headers: { 'x-api-key': 'itech_live_sec_7f9084e1b9384e9d9a1f' } }
  );
  console.log('WhatsApp Delivery:', res.data);
}
notifyDeviceReady();
```

8.3 PHP (Laravel, CodeIgniter, Core PHP cURL)

```php
<?php
$url = 'https://whatapp-automation-kxmi.onrender.com/api/v1/crm/repair-status-update';
$data = [
    'phone' => '919876543210',
    'customerName' => 'Ramesh Babu',
    'jobId' => 'REP-2026-081',
    'device' => 'Samsung S23',
    'status' => 'Screen Replacement Completed - Ready for Pickup'
];

$ch = curl_init($url);
curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
curl_setopt($ch, CURLOPT_POST, true);
curl_setopt($ch, CURLOPT_POSTFIELDS, json_encode($data));
curl_setopt($ch, CURLOPT_HTTPHEADER, [
    'Content-Type: application/json',
    'x-api-key: itech_live_sec_7f9084e1b9384e9d9a1f'
]);
$response = curl_exec($ch);
curl_close($ch);
echo $response;
```

8.4 C# / .NET (Desktop POS, WPF, ASP.NET Core)

```
using System.Net.Http;
using System.Text;
using System.Text.Json;

var client = new HttpClient();
client.DefaultRequestHeaders.Add("x-api-key", "itech_live_sec_7f9984e1b9304e9d9a1f");

var payload = new {
    phone = "919876543210",
    customerName = "Ramesh Babu",
    jobId = "REP-2026-001",
    device = "Samsung Galaxy S23",
    service = "Screen Glass Replacement",
    estDelivery = "Today by 6:00 PM"
};

var content = new StringContent(JsonSerializer.Serialize(payload), Encoding.UTF8, "application/json");
var response = await client.PostAsync("https://whatapp-automation-kxwl.onrender.com/api/v1/crm/repair-booking-confirmed", content);
var result = await response.Content.ReadAsStringAsync();
```

9. Error Codes, Rate Limits & Retry Policy

Standard HTTP response codes returned by the gateway and recommended client behavior.

Status Code	Meaning & Root Cause	Recommended ERP Action
200 OK	Message submitted to Meta and dispatched successfully.	Log success timestamp and reference ID in ERP.
400 Bad Request	Missing required fields (e.g. phone or customerName).	Validate phone number format before making request.
401 Unauthorized	Invalid or missing x-api-key header.	Check API secret key in your ERP settings.
429 Rate Limit	Meta Cloud API hourly threshold reached.	Retry with exponential backoff (e.g. 5, 15, 30 seconds).
500 Server Error	Gateway or database connectivity issue.	Enqueue in local ERP retry queue; retry after 10s.

KEY META WHATSAPP COMPLIANCE RULES:
• **24-Hour Session Freedom:** Standard free-text messages can only be sent if the customer messaged the business within the last 24 hours. However, all endpoints in this guide (repair confirmation, ready for pickup, dispatch alerts) use **pre-approved Meta Utility Templates**. They bypass this 24-hour limit and can be delivered **24/7/365**.
• **Phone Normalization:** The gateway automatically cleans spaces, hyphens, and leading zeros, and adds the country code (91 for India) if omitted.
• **Production Live Web Portal:** Test every endpoint live and view customer chat responses in real-time at: **https://itechmobileswhatsappautomation.vercel.app**

10. Postman Collection & Direct Technical Support

A complete, ready-to-import Postman Collection is available directly from the production server:

Postman Collection URL:	**https://whatapp-automation-kxml.onrender.com/itechnic-whatsapp-postman-collection.json**
OpenAPI 3.0 Spec:	**https://whatapp-automation-kxml.onrender.com/openapi.json**
Interactive Web Docs:	**https://itechmobileswhatsappautomation.vercel.app** (API Hub Tab)

TECHNICAL ASSISTANCE & INTEGRATION SUPPORT

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